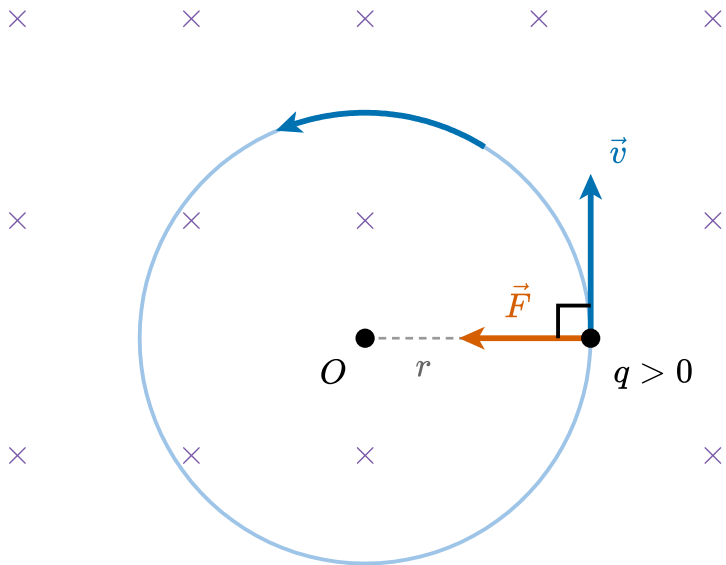


$\vec{B}$ into the page



$$\vec{F} = q \vec{v} \times \vec{B} \Rightarrow \|\vec{F}\| = |q|vB, \quad \vec{F} \perp \vec{v} \text{ automatically}$$

$$|q|vB = \frac{mv^2}{r} \implies r = \frac{mv}{|q|B}$$